

Java Assignment 4- Classes and Objects

1. Employee Salary Calculation

Create an `Employee` class with the following data members:

- Employee ID
- Employee Name
- Basic Salary
- HRA
- DA

Create the following methods:

- `read()` – to read employee details.
- `calculateSalary()` – to calculate the gross salary.
- `display()` – to display employee details and gross salary.

Formula:

`Gross Salary = Basic Salary + HRA + DA`

Create an object of the `Employee` class and display the result.

2. Bank Account Management

Create a `BankAccount` class with the following data members:

- Account Number
- Customer Name
- Balance

Create the following methods:

- `read()` – to read account details.
- `deposit()` – to deposit an amount into the account.
- `withdraw()` – to withdraw an amount if sufficient balance is available.
- `display()` – to display account details and current balance.

Create an object and perform **one deposit and one withdrawal operation**.

3. Product Billing

Create a `Product` class with the following data members:

- Product ID
- Product Name
- Price
- Quantity

Create the following methods:

- `read()` – to read product details.
- `calculateBill()` – to calculate the total amount.
- `display()` – to display product details and total bill amount.

Formula:

`Total Amount = Price × Quantity`

Create an object of the class and display the bill.

4. Electricity Bill Calculation

Create an `ElectricityBill` class with the following data members:

- Consumer Number
- Consumer Name
- Number of Units

Create the following methods:

- `read()` – to read consumer details and units consumed.
- `calculateBill()` – to calculate the electricity bill.
- `display()` – to display consumer details and bill amount.

Use the following tariff:

First 100 units	→ ₹2 per unit
Next 100 units	→ ₹3 per unit
Above 200 units	→ ₹5 per unit

Create an object and display the electricity bill.

5. Movie Ticket Booking

Create a `MovieTicket` class with the following data members:

- Customer Name
- Movie Name
- Number of Tickets
- Ticket Price

Create the following methods:

- `read()` – to read customer and ticket details.
- `calculateAmount()` – to calculate the total ticket amount.
- `display()` – to display booking details and total amount.

Formula:

`Total Amount = Number of Tickets × Ticket Price`